Light

GCSE Computer Science - Programming Data Structures (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A program needs to store the names of all 30 students in a class. What is the most efficient approach?

- ○ **A.** Create 30 separate string variables named student1, student2, student3, and so on
- ○ **B.** Store all names in a single string variable separated by commas
- ○ **C.** Use an array that can hold 30 string values
- ○ **D.** Use 30 separate integer variables

Q2 (1 mark)

What term describes the number used to identify the position of an element within an array?

- ○ **A.** The element's index
- ○ **B.** The element's field
- ○ **C.** The element's record
- ○ **D.** The element's key

Q3 (1 mark)

Which of the following best describes when you would use a record instead of an array?

- ○ **A.** When you need to store a list of values that are all the same type
- ○ **B.** When you need to group together different pieces of information about a single entity
- ○ **C.** When you need to perform mathematical calculations
- ○ **D.** When you need to store exactly two values

Q4 (1 mark)

Given an array `prices <- [9.99, 14.50, 7.25, 12.00]`, which expression retrieves the value 7.25?

- ○ **A.** prices[3]
- ○ **B.** prices[7.25]
- ○ **C.** prices[2]
- ○ **D.** prices(2)

Q5 (1 mark)

A noughts and crosses (tic-tac-toe) board has 3 rows and 3 columns. What type of data structure would best represent this board?

- ○ **A.** A record with 9 fields
- ○ **B.** A single one-dimensional array with 9 elements
- ○ **C.** A two-dimensional array with 3 rows and 3 columns
- ○ **D.** 9 separate variables

Q6 (1 mark)

A record called `movie` has fields `title`, `director`, and `year`. How do you access the director's name?

- ○ **A.** director.movie
- ○ **B.** movie[director]
- ○ **C.** movie.director
- ○ **D.** director(movie)

Q7 (1 mark)

An array is declared as `colours <- ["Red", "Blue", "Green", "Yellow", "Orange", "Purple"]`. What is the length of this array?

- ○ **A.** 6
- ○ **B.** 5
- ○ **C.** 7

- ○ **D.** 0

Q8 (1 mark)

In a record definition, why do we specify a data type for each field?

- ○ **A.** To make the program run faster
- ○ **B.** To limit the record to storing only one field
- ○ **C.** Data types are optional and only used for decoration
- ○ **D.** To ensure each field stores the appropriate kind of data and enable correct operations

Q9 (1 mark)

After executing `scores <- [85, 90, 78]` followed by `scores[1] <- 95`, what does the array contain?

- ○ **A.** [85, 90, 78]
- ○ **B.** [95, 90, 78]
- ○ **C.** [85, 95, 78]
- ○ **D.** [85, 90, 95]

Q10 (1 mark)

In a 2D array called `table`, what does the first index typically represent?

- ○ **A.** The value stored in the cell
- ○ **B.** The column position
- ○ **C.** The row position
- ○ **D.** The total number of elements

Q11 (1 mark)

After defining a record type `Student`, which statement creates an instance of that record?

- ○ **A.** pupil <- Student("Emma", 15, "10B")
- ○ **B.** Student <- RECORD
- ○ **C.** CREATE Student
- ○ **D.** DEFINE pupil AS Student

Q12 (1 mark)

What does an empty array contain?

- ○ **A.** The number zero
- ○ **B.** One empty element
- ○ **C.** No elements at all
- ○ **D.** Spaces filling the memory

Q13 (2 marks)

What is the value of `result` after this code executes?

```
data <- [12, 5, 8, 15, 3]
result <- data[4] + data[0]
```

- A. 17
- B. 27
- C. 20
- D. 15

Q14 (2 marks)

What is output by this code?

```
values <- [2, 4, 6]
product <- 1
FOR i <- 0 TO 2
    product <- product * values[i]
ENDFOR
OUTPUT product
```

- A. 48
- B. 12
- C. 24
- D. 6

Q15 (2 marks)

What is output by this code?

```
RECORD Vehicle
    make: String
    model: String
    mileage: Integer
ENDRECORD

car <- Vehicle("Toyota", "Corolla", 45000)
OUTPUT car.make
OUTPUT car.mileage
```

- ○ **A.** Toyota and Corolla
- ○ **B.** Toyota and 45000
- ○ **C.** Corolla and 45000
- ○ **D.** Vehicle and car

Q16 (2 marks)

A 2D array stores a multiplication table:

```
mult <- [[1, 2, 3],
         [2, 4, 6],
         [3, 6, 9]]
```

What is the value of `mult[1][2] + mult[2][1]`?

- ○ **A.** 10
- ○ **B.** 12
- ○ **C.** 8

- D. 15

Q17 (2 marks)

What is output by this code?

```
names <- ["Alice", "Bob", "Charlie", "Diana"]
found <- False
FOR i <- 0 TO 3
    IF names[i] = "Eve" THEN
        found <- True
    ENDIF
ENDFOR
OUTPUT found
```

- A. True
- B. 0
- C. Eve
- D. False

Q18 (2 marks)

What is output by this code?

```
RECORD Rectangle
    width: Integer
    height: Integer
ENDRECORD

box <- Rectangle(8, 5)
area <- box.width * box.height
OUTPUT area
```

- A. 13
- B. 40
- C. 85
- D. 3

Q19 (2 marks)

What is output by this code?

```
temps <- [15, 22, 28, 19, 31, 25]
hot <- 0
FOR i <- 0 TO 5
    IF temps[i] >= 25 THEN
        hot <- hot + 1
    ENDIF
ENDFOR
OUTPUT hot
```

- A. 3
- B. 2
- C. 4
- D. 6

Q20 (2 marks)

How many elements can be stored in a 2D array declared as `grid[4][5]`?

- A. 9

- B. 4
- C. 45
- D. 20

Q21 (2 marks)

An array has n elements. Using zero-based indexing, which expression gives the index of the last element?

- A. n
- B. n + 1
- C. n - 1
- D. 1

Q22 (2 marks)

What are the field values after this code executes?

```
RECORD Player
    name: String
    score: Integer
    lives: Integer
ENDRECORD

gamer <- Player("Alex", 0, 3)
gamer.score <- gamer.score + 100
gamer.lives <- gamer.lives - 1
```

- A. name="Alex", score=100, lives=2
- B. name="Alex", score=0, lives=3

- C. name="Alex", score=100, lives=3
- D. name="Alex", score=0, lives=2

Q23 (2 marks)

What is output by this code?

```
words <- ["Hello", "World"]
result <- words[0] + " " + words[1]
OUTPUT result
```

- A. HelloWorld
- B. Hello World
- C. Hello, World
- D. 01

Q24 (3 marks)

What is output by this code?

```
grid <- [[5, 10],
         [15, 20],
         [25, 30]]
max <- grid[0][0]
FOR r <- 0 TO 2
    FOR c <- 0 TO 1
        IF grid[r][c] > max THEN
            max <- grid[r][c]
        ENDIF
    ENDFOR
ENDFOR
OUTPUT max
```

- A. 5
- B. 20
- C. 25
- D. 30

Q25 (3 marks)

What is output by this code?

```
RECORD Item
    name: String
    price: Real
ENDRECORD

shop <- [Item("Pen", 1.50), Item("Book", 12.99), Item("Ruler", 0.
count <- 0
FOR i <- 0 TO 3
    IF shop[i].price > 10.00 THEN
        count <- count + 1
    ENDIF
ENDFOR
OUTPUT count
```

- A. 1
- B. 4
- C. 3
- D. 2

Q26 (3 marks)

What are the contents of the array after this code executes?

```
arr <- [40, 10, 30, 20]
FOR i <- 0 TO 2
    IF arr[i] > arr[i + 1] THEN
        temp <- arr[i]
        arr[i] <- arr[i + 1]
        arr[i + 1] <- temp
    ENDIF
ENDFOR
```

- ○ **A.** [40, 10, 30, 20]
- ○ **B.** [10, 30, 20, 40]
- ○ **C.** [10, 20, 30, 40]
- ○ **D.** [20, 30, 40, 10]

Q27 (3 marks)

What is output by this code that processes a square 2D array?

```
matrix <- [[7, 2, 9],
           [4, 5, 1],
           [8, 3, 6]]
total <- 0
FOR i <- 0 TO 2
    total <- total + matrix[i][i]
ENDFOR
OUTPUT total
```

- ○ **A.** 45
- ○ **B.** 24
- ○ **C.** 18

- ○ **D.** 15

Q28 (3 marks)

A hospital system needs to store patient information: NHS number (10-digit code), full name, date of birth, and whether they have any allergies. Analyse which record definition uses the most appropriate data types.

- ○ **A.** RECORD Patient nhsNumber: String fullName: String dateOfBirth: String hasAllergies: Boolean ENDRECORD
- ○ **B.** RECORD Patient nhsNumber: Integer fullName: Integer dateOfBirth: Integer hasAllergies: Integer ENDRECORD
- ○ **C.** RECORD Patient nhsNumber: Boolean fullName: Boolean dateOfBirth: Boolean hasAllergies: String ENDRECORD
- ○ **D.** RECORD Patient allInfo: String ENDRECORD

Q29 (3 marks)

What is output by this code that analyses an array?

```
votes <- ["A", "B", "A", "C", "A", "B", "A"]
countA <- 0
FOR i <- 0 TO 6
    IF votes[i] = "A" THEN
        countA <- countA + 1
    ENDIF
ENDFOR
IF countA > 3 THEN
    OUTPUT "A wins"
ELSE
    OUTPUT "No winner"
ENDIF
```

- A. No winner
- B. A
- C. 4
- D. A wins

Q30 (3 marks)

A 2D array stores test scores where rows are students and columns are subjects. Analyse what this code calculates.

```
scores <- [[70, 85, 90],
           [65, 75, 80],
           [95, 88, 92]]
rowTotals <- [0, 0, 0]
FOR r <- 0 TO 2
    FOR c <- 0 TO 2
        rowTotals[r] <- rowTotals[r] + scores[r][c]
    ENDFOR
ENDFOR
```

What does the array `rowTotals` contain after execution?

- A. [230, 248, 262]
- B. [245, 220, 275]
- C. [70, 65, 95]
- D. [90, 80, 92]

Total: **55** marks • Time: **30 min**

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